

ZHALTYR, KazakhstanWMO: **351730**Lat: **51.617N**Lon: **69.800E**Elev: **1001**StdP: **14.17**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.0	-22.0	-33.5	0.9	-26.7	-27.7	1.2	-21.7	26.9	18.1	22.6	16.1	4.6	50	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	90.2	63.9	86.7	63.4	83.4	62.7	67.3	82.4	66.0	81.1	64.6	78.9	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	88.0	71.6	60.9	82.9	70.8	59.3	78.2	69.9	32.3	82.4	31.3	81.5	30.2	78.8	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	18.7	16.8	DB	-32.1	95.7	5.1	3.6	-35.8	98.3	-38.8	100.4	-41.7	102.4	-45.5	105.0	(4)
(5)			WB	-32.1	69.5	5.1	1.6	-35.8	70.6	-38.8	71.6	-41.6	72.4	-45.3	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	1.2	4.5	18.8	42.9	56.9	65.9	67.6	65.2	54.4	40.7	21.9	8.3	
	DBStd	26.47	13.80	12.49	11.90	11.50	8.93	7.74	6.79	7.70	9.11	9.63	13.46	13.20	
	HDD50	6569	1511	1274	968	269	36	2	0	2	54	317	843	1293	
	HDD65	10412	1976	1694	1433	665	276	82	51	94	336	754	1293	1758	
	CDD50	2020	0	0	0	55	250	480	545	474	188	28	0	0	
	CDD65	388	0	0	0	1	26	110	131	101	19	0	0	0	
	CDH74	4566	0	0	0	22	388	1333	1407	1152	259	5	0	0	
	CDH80	1559	0	0	0	2	91	483	485	414	84	0	0	0	
Wind	WSAvg	7.1	7.1	7.9	8.0	7.8	7.6	6.9	6.1	6.0	6.3	6.8	7.4	7.7	
Precipitation	PrecAvg	13.1	0.8	0.7	0.8	0.8	1.2	1.6	2.2	1.3	0.8	0.9	0.9	1.0	
	PrecMax	21.0	1.4	1.7	1.8	1.8	3.9	3.9	6.0	3.2	2.1	2.4	2.2	2.2	
	PrecMin	7.9	0.1	0.2	0.2	0.1	0.1	0.4	0.4	0.3	0.1	0.2	0.3	0.2	
	PrecStd	2.8	0.4	0.4	0.5	0.4	0.8	0.9	1.2	0.7	0.6	0.5	0.5	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.9	33.5	48.7	76.9	87.4	93.8	94.7	93.3	89.4	73.1	49.2	33.7	
		MCWB	31.4	31.6	43.0	56.8	61.5	64.0	66.0	64.5	61.3	53.6	43.2	32.3	
	2%	DB	29.3	30.1	39.9	71.1	82.2	89.1	89.3	88.7	80.7	66.4	44.2	31.7	
		MCWB	28.1	28.3	36.7	55.2	59.1	63.0	65.2	63.7	58.3	51.1	40.7	30.5	
	5%	DB	25.2	26.8	36.8	66.3	78.2	85.7	85.2	84.8	75.3	61.0	40.3	29.0	
		MCWB	24.1	25.5	34.3	52.4	57.7	62.9	64.6	63.2	56.4	48.4	37.3	27.6	
	10%	DB	22.0	23.1	34.2	60.8	74.4	81.8	81.7	80.7	70.0	55.4	36.9	26.0	
		MCWB	20.8	21.9	32.3	49.2	56.0	61.5	63.8	61.7	54.0	46.0	34.4	24.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.7	31.7	43.4	59.4	64.6	68.0	69.3	68.0	62.8	55.2	44.5	32.8
			MCDB	32.7	33.2	48.6	72.5	81.0	83.3	83.8	83.9	84.3	68.6	47.7	33.4
2%		WB	28.3	28.4	37.2	56.0	61.2	66.2	67.8	66.3	59.8	52.0	41.1	30.5	
		MCDB	29.4	30.0	39.6	69.5	76.7	82.7	82.5	82.0	76.8	63.8	44.3	31.5	
5%		WB	24.2	25.4	34.5	53.2	59.3	64.6	66.5	64.7	57.5	49.2	37.4	27.7	
		MCDB	25.1	26.8	36.4	64.7	74.4	80.5	80.9	80.2	72.9	59.2	40.0	29.0	
10%		WB	20.9	22.0	32.3	50.2	57.3	63.0	65.2	63.1	55.3	46.4	34.4	24.8	
		MCDB	22.0	23.1	34.3	59.4	71.6	78.4	78.5	77.8	67.9	55.0	36.7	26.1	
Mean Daily Temperature Range		5% DB	MDBR	13.5	15.0	14.9	19.2	22.9	22.3	20.6	22.3	21.4	17.0	12.5	13.3
			MCDBR	12.4	13.8	12.0	26.8	28.0	27.0	25.5	27.3	28.1	25.8	14.0	13.4
	MCWBR		12.4	13.5	9.9	14.6	12.2	10.3	9.2	10.2	12.2	15.0	10.9	12.7	
	5% WB	MCDBR	12.4	13.9	11.7	25.4	25.0	23.7	22.1	24.2	26.0	23.2	13.2	12.9	
		MCWBR	12.4	13.6	10.0	14.7	12.7	10.2	9.4	10.5	12.1	14.4	10.8	12.5	
Clear-Sky Solar Irradiance	taub	0.292	0.322	0.374	0.407	0.400	0.406	0.425	0.410	0.358	0.342	0.324	0.292		
	taud	2.059	2.014	1.966	2.141	2.232	2.285	2.230	2.254	2.348	2.300	2.151	2.053		
	Ebn at Noon	215	243	253	262	270	269	261	259	261	239	202	191		
	Edh at Noon	25	36	46	43	42	40	42	38	31	27	23	22		
All-Sky Solar Radiation	RadAvg	434	796	1230	1518	1883	1998	1835	1619	1201	659	308	286		
	RadStd	32	33	127	138	167	165	144	113	87	86	35	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page